

Continuous-time Fourier Transform

Property	Signal	Fourier Transform
Linearity	$ax(t) + by(t)$	$aX(j\omega) + bY(j\omega)$
Convolution	$x(t) * y(t)$	$X(j\omega)Y(j\omega)$
Multiplication	$x(t)y(t)$	$\frac{1}{2\pi}X(j\omega) * Y(j\omega)$
Frequency Shifting	$e^{j\omega_0 t}x(t)$	$X(j(\omega - \omega_0))$
	$\cos(\omega_0 t)x(t)$	$\frac{1}{2}X(j(\omega - \omega_0)) + \frac{1}{2}X(j(\omega + \omega_0))$
Time Shifting	$x(t - t_0)$	$e^{-j\omega t_0}X(j\omega)$
Conjugation	$x^*(t)$	$X^*(-j\omega)$
	$x(t)$ real	$X(j\omega) = X^*(-j\omega)$
Differentiation in Time	$\frac{d}{dt}x(t)$	$j\omega X(j\omega)$
Differentiation in Frequency	$tx(t)$	$j\frac{d}{d\omega}X(j\omega)$
Integration	$\int_{-\infty}^t x(\lambda)d\lambda$	$\frac{1}{j\omega}X(j\omega) + \pi X(0)\delta(\omega)$
Scaling	$x(at), a \neq 0$	$\frac{1}{ a }X(j\frac{\omega}{a})$
Duality	$X(jt)$	$2\pi x(-\omega)$
Parseval		$\int_{-\infty}^{\infty} x(t) ^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) ^2 d\omega$

Signal	Fourier transform
$e^{-at}u(t), \text{Re}(a) > 0$	$\frac{1}{a+j\omega}$
$e^{-a t }$	$\frac{2a}{a^2+\omega^2}$
$te^{-at}u(t), \text{Re}(a) > 0$	$\frac{1}{(a+j\omega)^2}$
$u(t)$	$\frac{1}{j\omega} + \pi\delta(\omega)$
$\delta(t)$	1
$\delta(t - t_0)$	$e^{-j\omega t_0}$
1	$2\pi\delta(\omega)$
$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$
$\cos(\omega_0 t)$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
$\sin(\omega_0 t)$	$\frac{\pi}{j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$
$x(t) = \begin{cases} 1, & t \leq T_1 \\ 0, & t > T_1 \end{cases}$	$2\pi \frac{\sin(\omega T_1)}{\pi\omega}$
$\frac{\sin(Wt)}{\pi t}$	$X(j\omega) = \begin{cases} 1, & \omega < W \\ 0, & \omega > W \end{cases}$

Discrete Time Fourier Transform (DTFT) Properties and Pairs

Property	DT Signal	DT Fourier Transform [†]
Linearity	$ax[n] + by[n]$	$aX(e^{j\omega}) + bY(e^{j\omega})$
Convolution	$x[n] * y[n]$	$X(e^{j\omega})Y(e^{j\omega})$
Multiplication	$x[n]y[n]$	$\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\lambda})Y(e^{j(\omega-\lambda)})d\lambda$
Frequency Shifting	$e^{j\omega_0 n}x[n]$	$X(e^{j(\omega-\omega_0)})$
Time Shifting	$x[n-d]$	$e^{-j\omega d}X(e^{j\omega})$
Conjugation	$x^*[n]$	$X^*(e^{-j\omega})$
Time Reversal	$x[-n]$	$X(e^{-j\omega})$
Differentiation in Frequency	$n x[n]$	$j \frac{d}{d\omega} X(e^{j\omega})$
Conjugate symmetry	$x[n]$ real	$X(e^{j\omega}) = X^*(e^{-j\omega})$

[†] $X(e^{j\omega})$ is periodic with period 2π

DT Signal	DT Fourier transform [‡]
$a^n u[n], a < 1$	$\frac{1}{1 - ae^{-j\omega}}$
$(n+1)a^n u[n], a < 1$	$\frac{1}{(1 - ae^{-j\omega})^2}$
$u[n]$	$\frac{1}{1 - e^{-j\omega}} + \pi\delta(\omega)$
$\delta[n]$	1
$\delta[n-d]$	$e^{-j\omega d}$
1	$2\pi\delta(\omega)$
$e^{j\omega_0 n}$	$2\pi\delta(\omega - \omega_0)$
$\cos(\omega_0 n)$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
$\sin(\omega_0 n)$	$\frac{\pi}{j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$
$x[n] = \begin{cases} 1, & n \leq N_1 \\ 0, & n > N_1 \end{cases}$	$\frac{\sin(\omega(N_1 + 1/2))}{\sin(\omega/2)}$
$\frac{\sin(n\omega_c)}{\pi n}$	$X(e^{j\omega}) = \begin{cases} 1, & 0 \leq \omega \leq \omega_c \\ 0, & \omega_c \leq \omega \leq \pi \end{cases}$

[‡] $X(e^{j\omega})$ is periodic with period 2π and is listed here for one period only,
 $\omega \in [-\pi, \pi]$.

Unilateral Laplace Transform

Property	Time Domain	s -Domain
Independent Variable	t	s
Signal Representation	$x(t)$	$X(s)$
Linearity	$ax(t) + bv(t)$	$aX(s) + bV(s)$
Time shift	$x(t - c)u(t - c)$	$e^{-cs}X(s) \quad c > 0$
Time scaling	$x(at)$	$\frac{1}{a}X\left(\frac{s}{a}\right) \quad a > 0$
Multiply by t	$t^N x(t)$	$(-1)^N \frac{d^N}{ds^N} X(s), N = 1, 2, \dots$
Modulation	$e^{at}x(t)$	$X(s - a)$ (a real or complex)
	$x(t) \sin(\omega t)$	$\frac{j}{2} [X(s + j\omega) - X(s - j\omega)]$
	$x(t) \cos(\omega t)$	$\frac{1}{2} [X(s + j\omega) + X(s - j\omega)]$
Differentiation	$\frac{dx(t)}{dt}$	$sX(s) - x(0^-)$
	$\frac{d^2x(t)}{dt^2}$	$s^2X(s) - sx(0^-) - \left. \frac{dx(t)}{dt} \right _{t=0^-}$
Integration	$\int_0^t x(\tau) d\tau$	$\frac{1}{s}X(s)$
Convolution	$x(t) * v(t)$	$X(s)V(s)$
Initial Value Theorem	$\lim_{t \rightarrow 0^+} x(t)$	$\lim_{s \rightarrow \infty} sX(s)$
	$\lim_{t \rightarrow 0^+} \dot{x}(t)$	$\lim_{s \rightarrow \infty} [s^2X(s) - sx(0)]$
Final Value Theorem	$\lim_{t \rightarrow \infty} x(t)$	$\lim_{s \rightarrow 0} sX(s)$ (if t limit exists)

Signal	Waveform	Transform
Impulse	$\delta(t)$	1
Step	$u(t)$	$\frac{1}{s}$
Ramps, etc.	$t^N u(t)$	$\frac{N!}{s^{N+1}} \quad N = 1, 2, \dots$
Exponential	$e^{-bt} u(t)$	$\frac{1}{s+b}$
Damped Ramp, etc.	$t^N e^{-bt} u(t)$	$\frac{N!}{(s+b)^{N+1}} \quad N = 1, 2, \dots$
Sine	$\sin(\omega t) u(t)$	$\frac{\omega}{s^2 + \omega^2}$
Cosine	$\cos(\omega t) u(t)$	$\frac{s}{s^2 + \omega^2}$
Damped Sine	$e^{-bt} \sin(\omega t) u(t)$	$\frac{\omega}{(s+b)^2 + \omega^2}$
Damped Cosine	$e^{-bt} \cos(\omega t) u(t)$	$\frac{(s+b)}{(s+b)^2 + \omega^2}$
Ramped Sine	$t \sin(\omega t) u(t)$	$\frac{2\omega s}{(s^2 + \omega^2)^2}$
Ramped Cosine	$t \cos(\omega t) u(t)$	$\frac{s^2 - \omega^2}{(s^2 + \omega^2)^2}$
Simple Complex Poles	$2 k e^{-\alpha t} \cos(\beta t + \angle k) u(t)$	$\frac{k}{s + \alpha - j\beta} + \frac{k^*}{s + \alpha + j\beta}$
Double Complex Poles	$2 k te^{-\alpha t} \cos(\beta t + \angle k) u(t)$	$\frac{k}{(s + \alpha - j\beta)^2} + \frac{k^*}{(s + \alpha + j\beta)^2}$

Unilateral Z-Transform

$$X(z) = \sum_{n=0}^{\infty} x[n]z^{-n}$$

Property	Time Domain	z-Domain
Independent Variable	n (integer)	z (complex)
Signal Representation	$x[n]$	$X(z)$
Linearity	$ax[n] + bv[n]$	$aX(z) + bV(z)$
Delay	$y[n - 1]$	$z^{-1}Y(z) + y[-1]$
Convolution	$x[n] * h[n]$ (causal signals)	$X(z)H(z)$
Modulation	$e^{j\omega_0 n}x[n]$	$X(e^{-j\omega_0}z)$
Conjugation	$x^*[n]$	$X^*(z^*)$
z-Differentiation	$nx[n]$	$-z\frac{d}{dz}X(z)$

Signal	Waveform	Transform
Impulse	$\delta[n]$	1
Delay	$\delta[n - d]$	z^{-d} , $d > 0$ integer
Step	$u[n]$	$\frac{1}{1 - z^{-1}}$
Exponential	$a^n u[n]$	$\frac{1}{1 - az^{-1}}$, a complex
	$na^n u[n]$	$\frac{az^{-1}}{(1 - az^{-1})^2}$
Damped Cosine	$r^n \cos(\omega_0 n)u[n]$	$\frac{1 - r \cos(\omega_0)z^{-1}}{1 - 2r \cos(\omega_0)z^{-1} + r^2 z^{-2}}$, r real
Damped Sine	$r^n \sin(\omega_0 n)u[n]$	$\frac{r \sin(\omega_0)z^{-1}}{1 - 2r \cos(\omega_0)z^{-1} + r^2 z^{-2}}$, r real

The Z-transform provides algebraic methods of solution for discrete-time difference equations, in analogy with Laplace transform for continuous-time differential equations. The transform also gives us the *transfer function*, $H(z)$, and characterization of a discrete-time system in terms of poles and zeros. A causal system is *stable* if all poles are inside the unit circle in the z -plane. The *impulse response* and transfer function make a Z-transform pair, $h[n] \leftrightarrow H(z)$. The *frequency response* is the special case of the transfer function evaluated on the unit circle, $H(e^{j\omega})$.